Austin DeCicco

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Education

Rutgers University, New Brunswick, NJ
M.S. in Mathematics — GPA 4.0/4.0 *Expected May 2027*

Rutgers University, New Brunswick, NJ
B.A. in Mathematics — GPA 3.96/4.0, ***Summa Cum Laude*** *May 2023*

University of Maryland, College Park, MD
B.S. in Economics — Minor in Statistics *May 2022*

Research Interests

Quantitative finance; Stochastic investment models; Regime-switching models; Reinforcement Learning; Econometrics; Combinatorics.

Projects/Research

Independent Research in Financial Mathematics
Supervised by Prof. Kasper Larsen, Rutgers *Fall 2026 – In Progress*

- Proposed a model of how far a CEX price spike retraces once arbitrage with a constant-product AMM pool closes the gap, later adapted into the LiquidityPred class project; now scoping a master's thesis problem in financial mathematics.

LiquidityPred: Price-Impact Prediction
With Alex Goferman *Fall 2026 – In Progress*
Foundations of Financial Statistics and Risk Management & Introduction to Fintech (Prof. Bettaieb), Rutgers

- Predicts the size of a mid-cap stock's next 30-minute move given a simulated large buy or sell, from ten years of intraday data; random forest, XGBoost, and neural networks are compared to an NNLS benchmark in a dashboard.

FX Portfolio Simulator
With Neel Barve *Fall 2026 – In Progress*
Algorithmic Trading and Portfolio Management (Prof. Bettaieb), Rutgers

- Currency strategies tuned in-sample and tested out-of-sample feed an efficient-frontier optimizer in an interactive app.

Statistical Properties of Permutations and SYT Reflected by the Robinson–Schensted Algorithm
With Jike Liu *Spring 2026*
Experimental Mathematics (Prof. Zeilberger), Rutgers

- Paper, Maple code, and data; two integer sequences not in the OEIS.

Selected Coursework

Graduate (in progress, Fall 2026): Theory of Functions of a Real Variable, Abstract Algebra, Combinatorics, Algorithmic Trading and Portfolio Management, Financial Statistics and Risk Management.

Graduate (Rutgers, completed): Experimental Mathematics (A), Graph Theory (A), Introduction to Mathematical Physics (A), Probability in Discrete Mathematics (A).

Undergraduate (Rutgers / Maryland): Introduction to Stochastic Processes (A), Mathematical Theory of Statistics (A), Advanced Topics in Econometrics (A), Econometrics (A+), Game Theory (A).

Experience

Independent Generalist Investor

U.S. Equity Markets; Residential Property *May 2023 – Present*

- Invest in special situations and thematic equities, with the largest gains from a post-restructuring position in American Coastal Insurance (ACIC, formerly UIHC) and from AI and semiconductor stocks in the 2023–2024 rally.
- Direct the renovation of a residential property: manage contractors, capital improvements, and project accounting.

Independent Cryptocurrency Arbitrage

Ethereum and VeChain DeFi Ecosystems *2020 – 2022*

- Manually executed hundreds of cross-DEX arbitrage trades on Ethereum (Uniswap) and VeChain (Vexchange, Betterswap, and others) during the 2020–2022 DeFi cycle; the entry threshold scaled with capital and gas to clear fixed and proportional fees plus a buffer for expected spread decay over the $\sim$5-minute manual execution window.

Technical Skills

Programming: Python, Maple, Java, SAS, Stata, MATLAB, LaTeX, HTML, CSS.

Quantitative: stochastic processes, mathematical statistics, time-series and panel-data econometrics, regression and hypothesis testing, combinatorics, Monte Carlo and bootstrap methods.

Markets: 10+ years personal trading (equities, options, crypto); fundamental and special-situations equity analysis.

Service & Outreach

Volunteer Firefighter — Flemington Fire Department, NJ *May 2024 – Present*

Executive, College Park Student Investment Group

University of Maryland *2020 – 2021*

- Led group meetings, prepared market commentary, and taught peers equity-market and trading mechanics.